

Parallel in time integration of dynamical low rank approximation

Benjamin Carrel, Martin J. Gander, Bart Vandereycken

Section of mathematics, University of Geneva

I. Dynamical low-rank approximation

Problem statement: We are interested in solving a general *large time-dependent matrix differential equation* in the form

$$\begin{aligned}\dot{X}(t) &= F(X(t)) \quad t \in [0, T], \\ X(0) &= X_0 \in \mathbb{R}^{m \times n}.\end{aligned}$$

Definition: DLRA [Koch and Lubich, 2007]

The **dynamical low-rank approximation** solves the modified problem

$$\begin{aligned}\dot{Y}(t) &= \mathcal{P}_{Y(t)}[F(Y(t))], \quad t \in [0, T] \\ Y(0) &= Y_0 \in \mathcal{M}_r,\end{aligned}$$

where $\mathcal{M}_r$ is the set of rank r matrices and $\mathcal{P}_A : \mathbb{R}^{m \times n} \rightarrow \mathcal{T}_A \mathcal{M}_r$ is the orthogonal projection onto the tangent space of $\mathcal{M}_r$ in A .

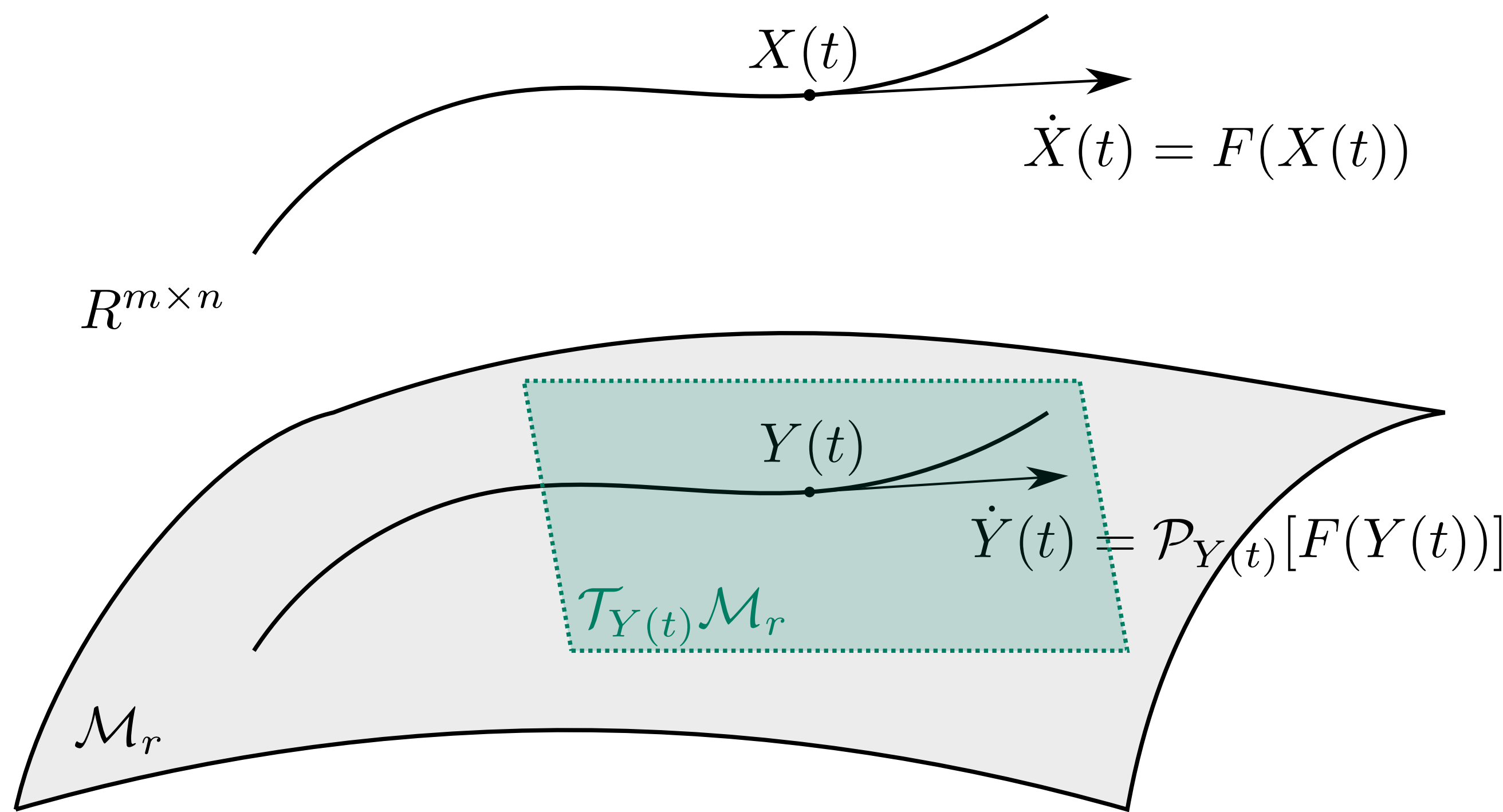


Figure 1: Visualisation of dynamical low-rank approximation.

Motivations:

- The solution $Y(t)$ of DLRA is *locally a good approximation* of the full rank solution $X(t)$.
- *Efficient storage and less computations* since $\text{rank}(Y(t)) = r \ll m, n$.

II. A parallel in time integrator

Algorithm: Parareal [Lions et al., 2001]

Suppose we have a *fine solver* $\mathcal{F}^h$ (expensive, accurate) and a *coarse solver* $\mathcal{G}^h$ (cheap, rough).

The **Parareal algorithm** iterates over $k, n = 0, \dots, N$,

$$\begin{aligned}(\text{Initial value}) \quad & U_0^k = X_0, \\ (\text{Initial iteration}) \quad & U_{n+1}^0 = \mathcal{G}^h(U_n^0), \\ (\text{Iteration}) \quad & U_{n+1}^{k+1} = \mathcal{F}^h(U_n^k) + \mathcal{G}^h(U_n^{k+1}) - \mathcal{G}^h(U_n^k),\end{aligned}$$

Properties:

- Easy to implement.
- Finite step convergence.
- Good performance on parabolic problems.

III. Our method

Key idea:

- Choose two ranks $q < r$.
- The **small rank** q defines a **coarse solver**.
- The **higher rank** r defines a **fine solver**.

Algorithm: Idealized low-rank Parareal [C., G., V., 2021]

$$\begin{aligned}(\text{Initial value}) \quad & X_0^k = \mathcal{T}_q(X_0) \\ (\text{Initial iteration}) \quad & X_{n+1}^0 = \phi^h \circ \mathcal{T}_q(X_n^0) \\ (\text{Iteration}) \quad & X_{n+1}^{k+1} = \phi^h \circ \mathcal{T}_r(X_n^k) + \phi^h \circ \mathcal{T}_q(X_n^{k+1}) - \phi^h \circ \mathcal{T}_q(X_n^k)\end{aligned}$$

where ϕ^h is the flow of the original problem, and $\mathcal{T}_q$ is the truncation to rank q .

Algorithm: Practical low-rank Parareal [C., G., V., 2021]

$$\begin{aligned}(\text{Initial value}) \quad & Y_0^k = Y_0, \\ (\text{Initial iteration}) \quad & Y_{n+1}^0 = \psi_q^h \circ \mathcal{T}_q(Y_n^0), \\ (\text{Iteration}) \quad & Y_{n+1}^{k+1} = \psi_r^h \circ \mathcal{T}_r(Y_n^k) + \psi_q^h \circ \mathcal{T}_q(Y_n^{k+1}) - \psi_q^h \circ \mathcal{T}_q(Y_n^k),\end{aligned}$$

where ψ_q^h is the flow of DLRA with rank q and $\mathcal{T}_q$ is the truncation to the rank q .

Remark: The rank of each iteration is at most $r + 2q$, so *all computations are low-rank*.

IV. Numerical example

Definition: Lyapunov differential equation

Let $A \in \mathbb{R}^{n \times n}$ be a square matrix and $B \in \mathbb{R}^{n \times k}$ be a tall matrix. The **Lyapunov differential equation** is

$$\dot{X} = AX + XA + BB^T, \quad X(0) = X_0.$$

Lyapunov equations have applications in *control theory* and *model reduction*.

When $A = \Delta$ is the 1-dimensional discrete Laplacian, this equation is a model for the **heat problem** where BB^T is the source.

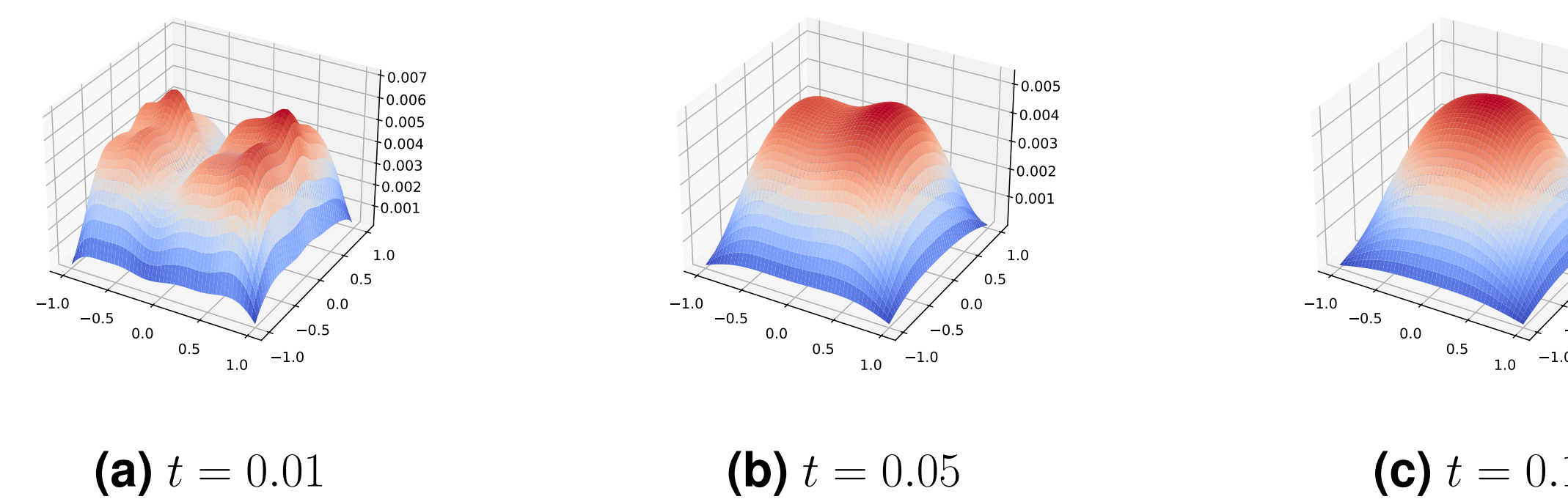


Figure 2: Solution over time.

Lemma: Low numerical rank of the solution [C., G., V., 2021]

The **numerical rank** is defined by

$$\text{rank}_\epsilon(A) = r \iff \min_{B \in \mathcal{M}_r} \|A - B\| \leq \epsilon.$$

The solution of **Lyapunov** is **low-rank** since

$$\text{rank}_{2\epsilon}(X(t)) \leq \text{rank}_\epsilon(X_0) + \text{rank}_\epsilon(B) \frac{\log(\epsilon/4) \log(4\kappa) - \log(e^{\lambda t} + 1)}{-\pi^2},$$

where $\lambda = \lambda_{\max}(A)$ and $\kappa = \kappa(A) = \|A\| \|A^{-1}\|$.

V. Convergence of the algorithm

The analysis holds only for the **idealized algorithm** applied to an **affine problem**. One can show the following error iteration,

$$\|X(t_{n+1}) - X_{n+1}^{k+1}\| \leq \alpha \|X(t_n) - X_n^k\| + \beta \|X(t_n) - X_n^{k+1}\|,$$

where α is the Lipschitz constant of $\phi^h \circ \mathcal{T}_q^\perp$, and β is the Lipschitz constant of $\phi^h \circ \mathcal{T}_q$.

Theorem: Convergence bound of idealized low-rank parareal [C., G., V., 2021]

Assume the problem is affine and $\beta < 1$. Then, the idealized algorithm verifies the **linear bound**

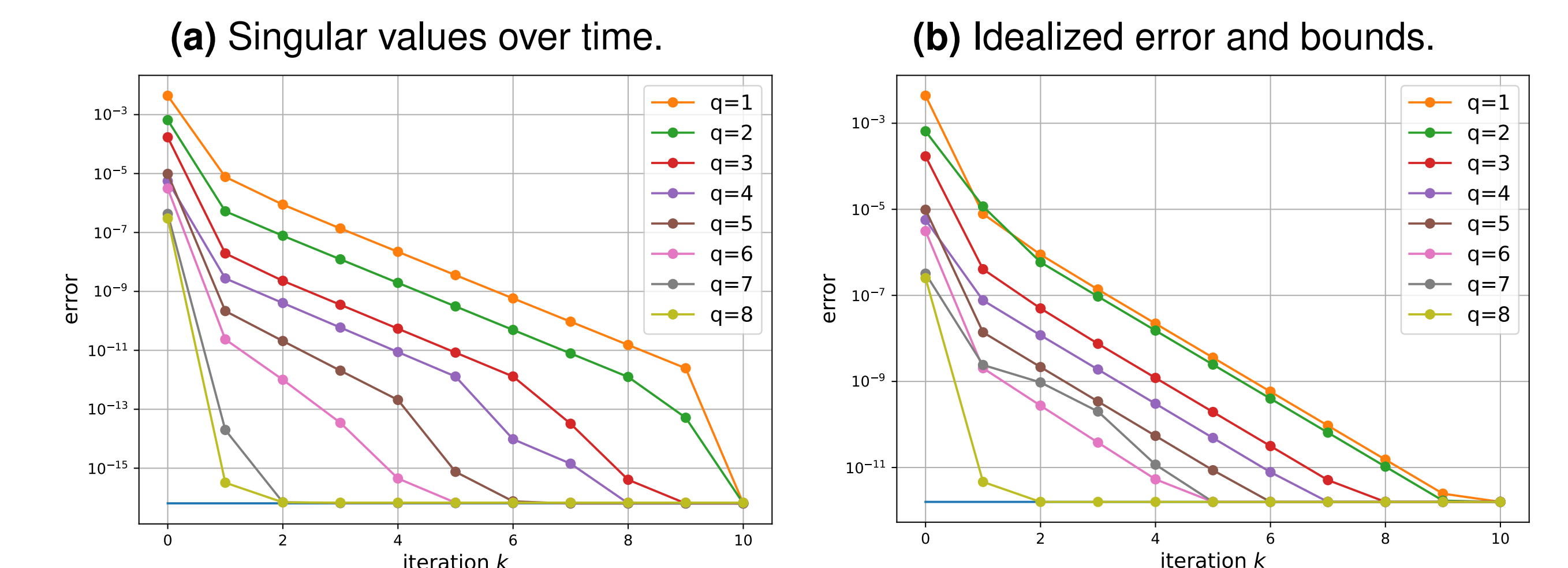
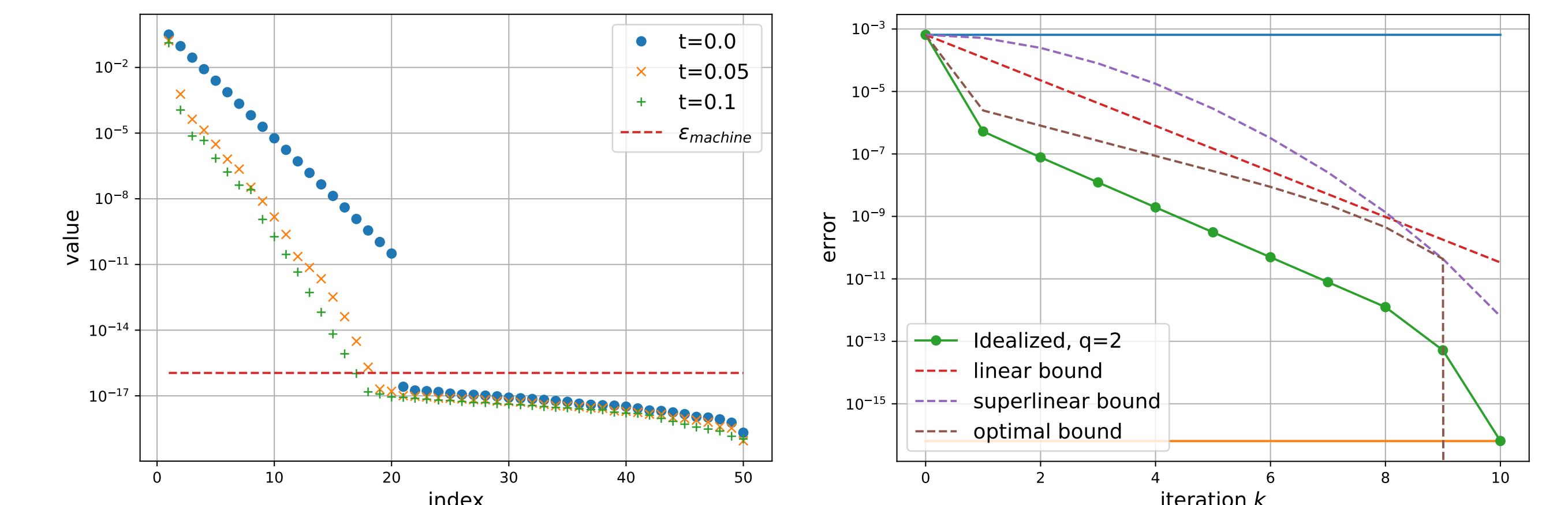
$$\max_{n \geq 0} \|X(t_n) - U_n^k\| \leq \frac{\alpha^k}{1 - \beta} \max_{n \geq 0} \|X(t_n) - U_n^0\|.$$

Moreover, it also verifies the **superlinear bound**

$$\max_{n \geq 0} \|X(t_n) - U_n^k\| \leq \frac{\alpha^k}{k!} \prod_{l=1}^k (N - l) \max_{n \geq 0} \|X(t_n) - U_n^0\|.$$

Finally, the best bound obtainable with the inequality given above is

$$\|X(t_n) - U_n^k\| \leq \alpha^k \sum_{i=0}^{n-k-1} \binom{k-1+i}{i} \beta^i \|X(t_{n-k-i}) - U_{n-k-i}^0\|.$$



(c) Idealized error, several ranks q .

(d) Practical error, several ranks q .

Figure 3: Idealized and practical low-rank Parareal errors.

References

- [Koch and Lubich, 2007] Koch, O. and Lubich, C. (2007). Dynamical Low-Rank Approximation. *SIAM J. Matrix Anal. & Appl.*, 29(2):434–454.
- [Lions et al., 2001] Lions, J.-L., Maday, Y., and Turinici, G. (2001). Résolution d'EDP par un schéma en temps pararéel. *Comptes Rendus de l'Académie des Sciences - Series I - Mathematics*, 332(7):661–668.